# A Galaxy in Disguise

Project Introduction | BubbleTea, Part I | Observational Astronomy

## Why look for a galaxy that resembles a star?

An ultra-compact dwarf galaxy (UCD) is a compact stellar system near the boundary between very massive star clusters and small galaxies. Its light comes from many stars, but at a large distance it can resemble a single star in an image. Some UCDs may be the surviving centers of larger galaxies stripped by gravitational interactions; others may have formed as unusually massive star clusters. Their origins are part of the research question, not settled by their name.

These small objects can reveal a much larger history. Detailed observations of M60-UCD1 found a massive black hole, supporting an origin in a larger galaxy. The evidence went far beyond its appearance in a photograph. Seth et al. (2014)

The UCD census remains incomplete. Much of the search effort has concentrated on galaxy clusters and groups. Could less-studied massive galaxies host overlooked UCDs? How common are they around disk galaxies, or in less crowded environments? These are motivations for this semester's project, not a promise that any particular galaxy will contain a detectable UCD or that we will make a new discovery. The HTML lesson provides examples and further reading.

## Two complementary views of the sky

Gaia repeatedly scanned the sky, recording positions, brightnesses, colors, parallaxes, and proper motions. It detected compact concentrations of light, not only objects already known to be stars. Compact galaxies and quasars can therefore also have Gaia records. Small detector pixels, the ability to separate two sources, and precision in measuring a position are different concepts. ESA: Gaia

Legacy Survey images provide a complementary view of an object's appearance and surroundings. Atmospheric and instrumental blur can make a small object look point-like. The same physical object also looks smaller at greater distance. A color picture is a display, not a substitute for measured fluxes or colors. This activity combines saved Gaia DR3 records and Legacy Survey DR11 images. Legacy Survey DR11

## Getting started

Save ucd_investigation.html to your computer and open it in a desktop browser, not an email or course-platform preview. All required data, images, and interactive tools are inside that one file. No coding, installation, account, local server, or internet connection is required for the core tasks. Optional online exploration requires internet access.

Begin with the background, illustrations, measurement definitions, and labeled examples. The core reading is designed for about 30 minutes; allow additional time for the investigation. This is Part I of a semester project. Searching for new candidates comes later; it is not a requirement of this assignment.

# Your investigation

The aim is to connect observations with physical interpretations, test a selection rule on real data, and explain both its usefulness and its limits. Use the examples to develop your own reasoning rather than seek a perfect shortcut. Definitions and optional tool instructions are available in the HTML.

## Task 1 | Classify 30 objects

Inspect each object's close and wider images, Gaia measurements, and available links. Choose UCD, star, QSO (quasar), or uncertain for every object. Only the classification is required. Confidence, evidence used, a short explanation, and external citations are optional but useful records of your reasoning. You can revisit and change your choices; the activity does not reveal the answers.

Read values together with their uncertainties. “Not supplied” is not zero. Distinguish what the observations show from what you infer. If you consult an online catalog, check that you are looking at the same source. The expandable guide explains Gaia Archive, Legacy Survey, and Aladin exploration; online checks are optional, not extra required deliverables.

## Task 2 | Design and improve a selection rule

Explore the interactive plots of 21,118 real catalog objects in a deliberately star-rich sample. Enable measurement ranges, adjust their limits, and decide how to handle missing measurements. Every enabled condition must be satisfied. Press Evaluate selection to record a trial and see its selected objects and scores; changing a slider alone does not evaluate a new trial.

- Completeness: the fraction of reference UCDs recovered by your rule.
- Purity: among selected objects with secure reference classes, the fraction that are UCDs rather than stars or QSOs.
- F1: a combined score, the harmonic mean of purity and completeness. Try to improve it while inspecting the counts and the two component scores.

Unconfirmed candidates are counted separately and do not enter these scores. The sample is constructed for teaching: scores after repeated tuning do not establish performance in a new sky field. A high score does not confirm an individual candidate as a UCD. Perfect separation is not the expectation.

Try different rules, compare the recorded attempts, then choose one evaluated attempt as your final selection. Complete both explanations in the page:

- Why do you think this works?
- How do you think this selection can be improved?

## Task 3 | Establishing membership

Answer the following question in the page. Your response will provide a starting point for Part II:

You find a possible UCD close to a massive galaxy on the sky. How would you investigate whether it is physically associated with that galaxy?

# What to deliver

Submit one JSON answer file exported from the HTML activity. It contains your Student ID, the 30 Task 1 responses, Task 2 trial history and final selection with both explanations, the Task 3 answer, and your brief LLM-use disclosure. There is no separate report, screenshot collection, code, or discovery claim required for Part I. The optional TXT export is for convenient reading and does not replace the JSON submission.

## Save, check, and submit

1. Under Keep a record, enter your Student ID. Complete the LLM-use declaration. If you used an LLM, briefly say how it helped and what you verified.
2. Check that all 30 objects have a classification, one evaluated Task 2 attempt is chosen as final, both selection explanations are filled in, and Task 3 has an answer. “Uncertain” is a valid classification, not a blank response.
3. Click Download answers · JSON. Find the downloaded file and rename it to [NAME]_[STUDENT_ID].json. Replace NAME with your name and STUDENT_ID with your student ID; remove the square brackets and retain .json.
4. Upload that JSON file to 网络学堂 (Tsinghua Online Classroom) in the assignment specified by your instructor. Check that the upload succeeded. Follow the course announcement for the deadline.
5. Keep a local copy of your final JSON and the exact HTML used to create it.

The webpage asks only for Student ID; your name belongs in the submission filename. On systems that hide file extensions, avoid accidentally adding .json twice. Downloading the file does not upload it to 网络学堂.

## Keep your work safe

Download JSON early and after important changes, even if your work is incomplete. Browser draft storage is convenient but not a dependable backup; moving the HTML or changing browsers may make the draft unavailable. The downloaded HTML itself does not acquire your answers.

To continue later, open the same HTML and choose your saved JSON under Resume from a saved JSON file. Import replaces the current answers, so export any work you want to retain first. Renaming the JSON does not alter its contents. Do not edit the JSON by hand. After further changes, download a new JSON and submit that final version, not an earlier draft.

## Responsible use of LLMs

LLM assistance is allowed and encouraged when used thoughtfully. It can help explain a concept, organize an investigation, or suggest checks, but its confident wording is not observational evidence. Verify factual claims against the supplied measurements and reliable sources. Your submitted choices and explanations should reflect what you actually checked. Provide a brief disclosure; no private chat transcript is requested.

The goal: a reproducible record of your decisions, a tested selection rule, and a thoughtful question about physical association. Those are the foundations for the next stage of the project.